

1. Create training

localhost:8080/swagger-ui/index.html#

POST /api/trainings Add training session

Parameters Cancel Reset

No parameters

Request body required application/json

```
{
  "traineeUsername": "Alice.Johnson",
  "trainerUsername": "Maria.Garcia",
  "trainingName": "Morning Full Body Yoga",
  "trainingDate": "2026-05-05",
  "duration": 50
}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  http://localhost:8080/api/trainings \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJ3bGJjZS5kb2huc29uIiwicm99ZS516lJ3PTEVFVFjBSUSFRSIsIm1hdCI6MTc3NzYyNzZMZWlZxhwIjoxNzc3NzEzZ2FQ.XtJ9-fq2ennkJEjUBd4Ywacfd5A-1XH-p04' \
  -H 'Content-Type: application/json' \
  -d '{
    "traineeUsername": "Alice.Johnson",
    "trainerUsername": "Maria.Garcia",
    "trainingName": "Morning Full Body Yoga",
    "trainingDate": "2026-05-05",
    "duration": 50
  }'
```

Request URL

http://localhost:8080/api/trainings

Server response

Code	Details
200	Response headers <pre>cache-control: no-cache,no-store,max-age=0,must-revalidate connection: keep-alive content-length: 0 date: Fri, 01 May 2026 10:12:56 GMT expires: 0 keep-alive: timeout=60 pragma: no-cache vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers x-content-type-options: nosniff x-frame-options: DENY x-transaction-id: 4579fed9-c5c0-4e8a-9140-7696330777b1 x-xss-protection: 0</pre>

Logging:

```
veronika.khan@Veronikas-MacBook-Pro gym-app % docker compose logs gym-core trainer-report-service | grep "4579fed9-c5c0-4e8a-9140-7696330777b1"
gym-core-1 | 2026-05-01 10:12:56 INFO [4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.c.f.TransactionIdFilter - >>> POST /api/trainings [client=140.8
2.121.33]
gym-core-1 | 2026-05-01 10:12:56 DEBUG [4579fed9-c5c0-4e8a-9140-7696330777b1] o.s.w.s.DispatcherServlet - POST "/api/trainings", parameters={}
gym-core-1 | 2026-05-01 10:12:56 DEBUG [4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.c.a.LoggingAspect - Calling: GymFacade.createTraining(...) args:
[Alice.Johnson, Maria.Garcia, Morning Full Body Yoga, 2026-05-05, 50]
gym-core-1 | 2026-05-01 10:12:56 DEBUG [4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.c.a.LoggingAspect - Calling: TraineeService.getTraineeByUsernam
e(...) args: [Alice.Johnson]
gym-core-1 | 2026-05-01 10:12:56 DEBUG [4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.c.a.LoggingAspect - Completed: TraineeService.getTraineeByUsernam
e(...)
gym-core-1 | 2026-05-01 10:12:56 DEBUG [4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.c.a.LoggingAspect - Calling: TrainerService.getTrainerByUsernam
e(...) args: [Maria.Garcia]
gym-core-1 | 2026-05-01 10:12:56 DEBUG [4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.c.a.LoggingAspect - Completed: TrainerService.getTrainerByUsernam
e(...)
gym-core-1 | 2026-05-01 10:12:56 DEBUG [4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.c.a.LoggingAspect - Calling: TrainingService.createTraining(...)
args: [Trainee(id=00000000-0000-0000-0000-000000000020, dateOfBirth=1995-03-14, address=12 Elm Street), Trainer(id=00000000-0000-0000-0000-000000000031), Morning
Full Body Yoga, TrainingType(id=00000000-0000-0000-0000-000000000002, trainingTypeName=Yoga), 2026-05-05, 50]
gym-core-1 | 2026-05-01 10:12:56 DEBUG [4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.c.a.LoggingAspect - Completed: TrainingService.createTraining(.
.)
gym-core-1 | 2026-05-01 10:12:56 DEBUG [4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.c.m.TrainingEventPublisher - [txId=4579fed9-c5c0-4e8a-9140-7696
330777b1] Published ADD event for trainer=Maria.Garcia
gym-core-1 | 2026-05-01 10:12:56 DEBUG [4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.c.a.LoggingAspect - Completed: GymFacade.createTraining(...)
gym-core-1 | 2026-05-01 10:12:56 DEBUG [4579fed9-c5c0-4e8a-9140-7696330777b1] o.s.w.s.DispatcherServlet - Completed 200 OK
gym-core-1 | 2026-05-01 10:12:56 INFO [4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.c.f.TransactionIdFilter - <<< POST /api/trainings -> HTTP 200
trainer-report-service-1 | 2026-05-01 10:12:56 DEBUG [txId=4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.r.m.TrainingEventListener - [txId=4579fed9-c5c0-4e8a-9140-
7696330777b1] Received ADD event for trainer=Maria.Garcia
trainer-report-service-1 | 2026-05-01 10:12:56 DEBUG [txId=4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.r.a.LoggingAspect - Calling: ReportService.processWorkLoad
(...) args: [com.epam.gym.report.dto.WorkLoadRequest@3635b055]
trainer-report-service-1 | 2026-05-01 10:12:56 INFO [txId=4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.r.s.ReportService - [txId=4579fed9-c5c0-4e8a-9140-76963307
77b1] processWorkLoad START action=ADD trainer=Maria.Garcia
trainer-report-service-1 | 2026-05-01 10:12:56 DEBUG [txId=4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.r.s.ReportService - [txId=4579fed9-c5c0-4e8a-9140-76963307
77b1] No existing record for trainer=Maria.Garcia, creating new document
trainer-report-service-1 | 2026-05-01 10:12:56 DEBUG [txId=4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.r.s.ReportService - [txId=4579fed9-c5c0-4e8a-9140-76963307
77b1] Creating year entry year=2026
trainer-report-service-1 | 2026-05-01 10:12:56 DEBUG [txId=4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.r.s.ReportService - [txId=4579fed9-c5c0-4e8a-9140-76963307
77b1] Creating month entry month=5
trainer-report-service-1 | 2026-05-01 10:12:56 DEBUG [txId=4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.r.s.ReportService - [txId=4579fed9-c5c0-4e8a-9140-76963307
77b1] duration update: year=2026 month=5 prev=0 deltas=50 next=50
trainer-report-service-1 | 2026-05-01 10:12:56 INFO [txId=4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.r.s.ReportService - [txId=4579fed9-c5c0-4e8a-9140-76963307
77b1] processWorkLoad END trainer=Maria.Garcia year=2026 month=5 duration=50
trainer-report-service-1 | 2026-05-01 10:12:56 DEBUG [txId=4579fed9-c5c0-4e8a-9140-7696330777b1] c.e.g.r.a.LoggingAspect - Completed: ReportService.processWorkLo
ad(...)
```

MongoDB:

The screenshot shows the MongoDB Compass application interface. The title bar indicates the connection is to 'localhost:27017/trainer_report.trainer_summary'. The left sidebar shows the 'Connections' list with 'localhost:27017' selected, and the 'Database Explorer' showing the 'trainer_report' database with the 'trainer_summary' collection highlighted. The main panel displays the 'Documents' tab for the 'trainer_summary' collection. A single document is shown with the following structure:

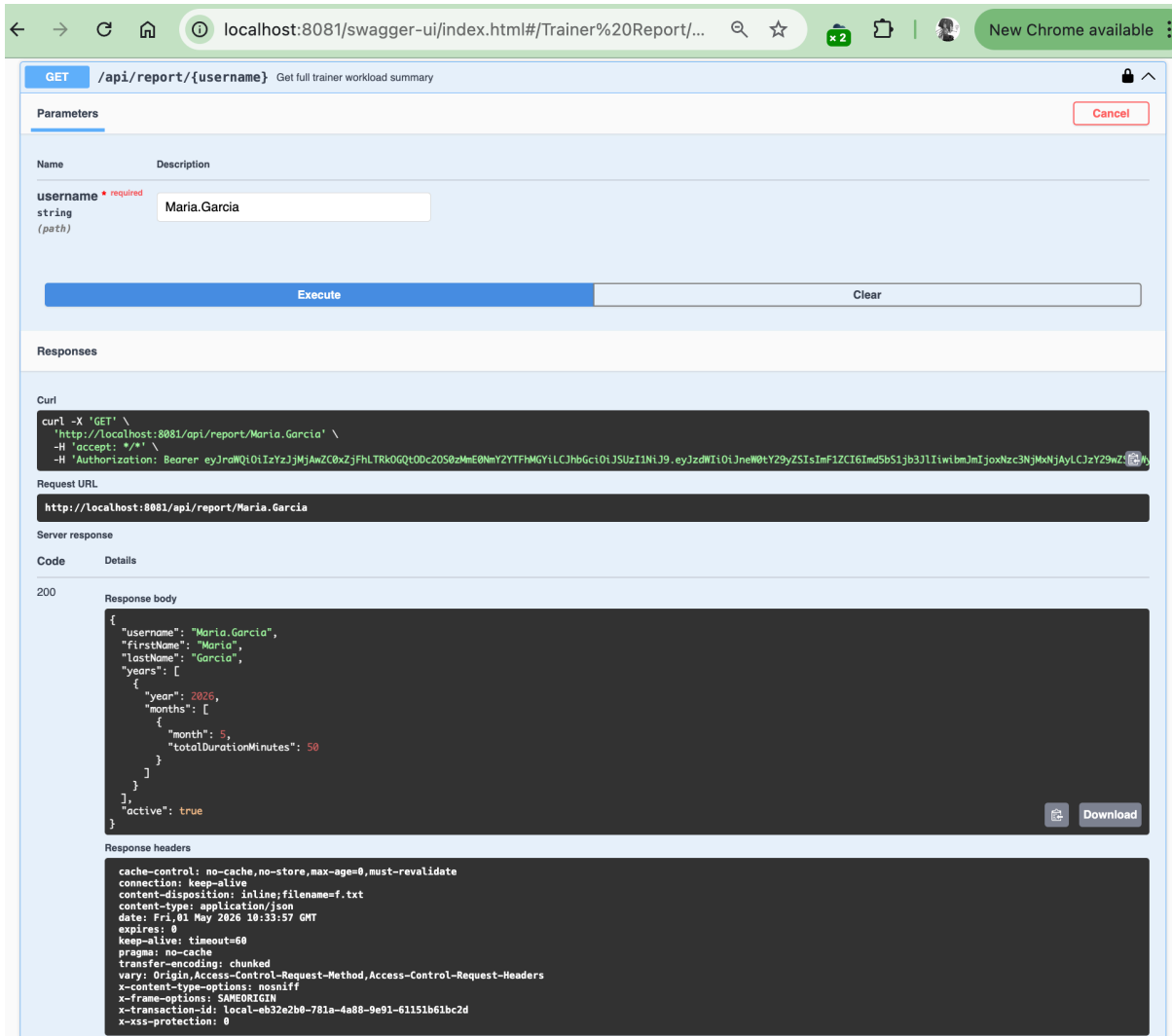
```
{
  "_id": ObjectId('69f47ca8f4b4bc7625e227f1'),
  "username": "Maria.Garcia",
  "firstName": "Maria",
  "lastName": "Garcia",
  "active": true,
  "years": [
    {
      "year": 2026,
      "months": [
        {
          "month": 5,
          "totalDurationMinutes": 50
        }
      ]
    }
  ]
}
```

The document is of type 'com.epam.gym.report.model.TrainerSummary'.

2. Check if index for last and first name is used (with query in mongosh):

```
trainer_report> db.trainer_summary.find({ lastName: "Garcia", firstName: "Maria" }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'trainer_report.trainer_summary',
    indexFilterSet: false,
    parsedQuery: {
      '$and': [
        { firstName: { '$eq': 'Maria' } },
        { lastName: { '$eq': 'Garcia' } }
      ]
    },
    queryHash: '41E425E3',
    planCacheKey: '42D44474',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    winningPlan: {
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { lastName: 1, firstName: 1 },
        indexName: 'idx_last_first_name',
        isMultiKey: false,
        multiKeyPaths: { lastName: [], firstName: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: {
          lastName: [ ["Garcia", "Garcia"] ],
          firstName: [ ["Maria", "Maria"] ]
        }
      }
    },
    rejectedPlans: []
  },
}
```

3. Get trainer's report by username:



Swagger UI interface for the `/api/report/{username}` endpoint. The interface shows the method `GET`, the endpoint path, and a parameter `username` with the value `Maria.Garcia`. Below the parameters, there are buttons for `Execute` and `Clear`. The `Responses` section shows a `200` status code with a JSON response body and response headers.

Parameters

Name	Description
<code>username</code> * required	
string	
(path)	

Execute **Clear**

Responses

Code **Details**

200

Response body

```
{
  "username": "Maria.Garcia",
  "firstName": "Maria",
  "lastName": "Garcia",
  "years": [
    {
      "year": 2026,
      "months": [
        {
          "month": 5,
          "totalDurationMinutes": 50
        }
      ]
    }
  ],
  "active": true
}
```

Response headers

```
cache-control: no-cache, no-store, max-age=0, must-revalidate
connection: keep-alive
content-disposition: inline; filename=f.txt
content-type: application/json
date: Fri, 01 May 2026 10:33:57 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
transfer-encoding: chunked
vary: Origin, Access-Control-Request-Method, Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: SAMEORIGIN
x-transaction-id: local-eb32e2b0-781a-4a88-9e91-61151b61bc2d
x-xss-protection: 0
```

Logging:

```
veronika.khan@Veronikas-MacBook-Pro gym-app % docker compose logs gym-core trainer-report-service | grep "local-eb32e2b0-781a-4a88-9e91-61151b61bc2d"
trainer-report-service-1 | 2026-05-01 10:33:57 INFO [txId=local-eb32e2b0-781a-4a88-9e91-61151b61bc2d] c.e.g.r.f.TransactionIdFilter - >>> GET /api/report/Maria.Garcia
trainer-report-service-1 | 2026-05-01 10:33:57 DEBUG [txId=local-eb32e2b0-781a-4a88-9e91-61151b61bc2d] o.s.w.s.DispatcherServlet - GET "/api/report/Maria.Garcia", parameters={}
trainer-report-service-1 | 2026-05-01 10:33:57 DEBUG [txId=local-eb32e2b0-781a-4a88-9e91-61151b61bc2d] c.e.g.r.a.LoggingAspect - Calling: ReportService.getSummary(..) args: [Maria.Garcia]
trainer-report-service-1 | 2026-05-01 10:33:57 INFO [txId=local-eb32e2b0-781a-4a88-9e91-61151b61bc2d] c.e.g.r.s.ReportService - [txId=local-eb32e2b0-781a-4a88-9e91-61151b61bc2d] getSummary username=Maria.Garcia
trainer-report-service-1 | 2026-05-01 10:33:57 DEBUG [txId=local-eb32e2b0-781a-4a88-9e91-61151b61bc2d] c.e.g.r.a.LoggingAspect - Completed: ReportService.getSummary(..)
trainer-report-service-1 | 2026-05-01 10:33:57 DEBUG [txId=local-eb32e2b0-781a-4a88-9e91-61151b61bc2d] o.s.w.s.DispatcherServlet - Completed 200 OK
trainer-report-service-1 | 2026-05-01 10:33:57 INFO [txId=local-eb32e2b0-781a-4a88-9e91-61151b61bc2d] c.e.g.r.f.TransactionIdFilter - <<< GET /api/report/Maria.Garcia ->> HTTP 200
```

4. Create new training for the same year and month:

The image shows the Swagger UI interface for a REST API. The URL is `localhost:8080/swagger-ui/index.html#`. The endpoint is `POST /api/trainings` with the description "Add training session". There are no parameters. The request body is required and set to `application/json`. The JSON body is:

```
{
  "traineeUsername": "Alice.Johnson",
  "trainerUsername": "Maria.Garcia",
  "trainingName": "Evening Stretching Yoga",
  "trainingDate": "2026-05-06",
  "duration": 50
}
```

The "Execute" button is visible. Below the request body, the "Responses" section shows a `200` status code with the following response headers:

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
connection: keep-alive
content-length: 0
date: Fri, 01 May 2026 10:37:52 GMT
expires: 0
keep-alive: timeout=60
pragma: no-cache
vary: Origin,Access-Control-Request-Method,Access-Control-Request-Headers
x-content-type-options: nosniff
x-frame-options: DENY
x-transaction-id: e14be8ae-a5fe-4e3f-90fd-8f142b105bb1
x-xss-protection: 0
```

Make sure changes reflected in MongoDB:

The image shows the MongoDB Compass interface. The database is `localhost:27017/trainer_report` and the collection is `trainer_summary`. The document is displayed in the "Documents" tab. The document structure is as follows:

```
{
  "_id": ObjectId('69f47ca8f4b4bc7625e227f1'),
  "username": "Maria.Garcia",
  "firstName": "Maria",
  "lastName": "Garcia",
  "active": true,
  "years": [
    {
      "year": 2026,
      "months": [
        {
          "month": 5,
          "totalDurationMinutes": 100
        }
      ]
    }
  ],
  "_class": "com.epam.gym.report.model.TrainerSummary"
}
```